

FoodFlow — Project Proposal | CIT6224

WEB APPLICATION DEVELOPMENT

CIT6224

Group 19

Project Title: FoodFlow

Student ID	Student Name
241UC240P4	Al-Maweri Yousef Mohammed Abdullah
241UC2400T	Bara Samih Jamal Yousef

Contents

Problem Definition and Business Context	3
Overview	3
Product and Service Scope	3
The Customer Journey.....	4
Technology and Platform Architecture	5
Business Differentiation	5
Identify Target Users and Analyse Their Needs.....	6
User Pain Points Analysis	6
Competitor Comparison	7
Discuss Existing Challenges or Gaps the Proposed System Aims to Address	7
Business Value of the Proposed System	8
Proposed System Solution	8
Define System Objectives and Key Functionalities	8
Non-Functional Requirements.....	11
System Workflow	12
How the Proposed System Improves Business Operations and User Experience	12
Future Enhancements.....	13
System Design and Modelling	13
Develop System Design Using Wireframes or User Interface Mockups	13
UI/UX Design Principles	14
Home Page Design	14
Restaurant & Menu Page Design	15
Shopping Cart Page Design	15
Login and Registration Page Design	16
Admin Dashboard Design.....	16
Summary.....	17
Project Planning and Management	17
Define Roles and Responsibilities of Each Group Member	17
Development Methodology	19
Project Risk Management.....	19
Identify Tools, Frameworks, and Technologies to Be Used	19
Project Timeline.....	20

Problem Definition and Business Context

Overview

FoodFlow is a Business-to-Consumer (B2C) food delivery web application developed for the online ordering and delivery of meals from local restaurants and food vendors. The platform provides customers with a unified digital environment where they can browse restaurant menus, search for dishes, place orders, and track deliveries in real time through a dedicated branded storefront. The system focuses on delivering a seamless and secure food ordering experience while simplifying order and menu management for restaurant partners and platform administrators.

Unlike open food aggregator platforms where pricing and branding are dictated by third-party operators, FoodFlow adopts a managed multi-vendor model. In this model, FoodFlow acts as the platform operator responsible for partner onboarding, quality control, and customer experience, while restaurant partners retain control over their menus and pricing. This allows the business to ensure consistent service standards, transparent pricing, and a reliable end-to-end delivery experience.

The system is developed using HTML5, CSS3, JavaScript, PHP, and MySQL within a XAMPP local server environment. The architecture separates frontend presentation from backend business logic to improve maintainability, scalability, and future system enhancement.

FoodFlow aims to provide a modern and responsive food delivery platform that supports customer convenience, restaurant partner operations, and efficient delivery coordination. The platform focuses specifically on local food vendors across multiple cuisine categories including local cuisine, western food, beverages, healthy meals, and grocery convenience items.

Product and Service Scope

FoodFlow's restaurant and menu catalogue is organised into five primary food categories, each catering to a distinct segment of the food delivery market:

Table 1 Food Category Catalogue

Category	Menu Examples	Target Segment
Local Cuisine	Nasi Lemak, Mee Goreng, Char Kuey Teow, Laksa	Students, working adults, families
Western & Fast Food	Burgers, pasta, fried chicken, sandwiches, wraps	Young adults, office workers, teenagers
Beverages & Desserts	Bubble tea, smoothies, cakes, ice cream, waffles	Students, health enthusiasts, casual snackers
Healthy & Diet Meals	Salads, grain bowls, low-calorie sets, vegan options	Fitness enthusiasts, health-conscious consumers
Snacks & Groceries	Instant noodles, chips, canned goods, condiments	Households, students, convenience shoppers

Revenue and Operational Model

FoodFlow operates using a commission-based business model where revenue is generated through a percentage service fee applied to each completed order from restaurant partners. The platform also charges a customer delivery fee that varies by distance and order size.

The business model does not depend on:

- paid advertisements or sponsored restaurant placements,
- subscription fees charged to customers,
- or fixed monthly fees charged to restaurant partners.

This allows the platform to align its revenue directly with transaction volume, encouraging a focus on platform reliability and customer satisfaction.

The system supports four primary user roles:

Customers

Customers can:

- browse restaurants and menus,
- search and filter by cuisine, price, or rating,
- manage shopping carts and add-ons,
- place and pay for orders,
- and track delivery status in real time.

Restaurant Partners

Restaurant partners manage their operational side through a dedicated partner dashboard.

They can:

- add, edit, and delete menu items,
- manage stock and item availability,
- accept or reject incoming orders,
- and view daily sales and revenue summaries.

Delivery Riders

Delivery riders receive job assignments and manage their active deliveries through a mobile-friendly interface. They can update order status at each stage of the delivery process.

Administrators

Administrators manage the entire platform through a dedicated admin panel. They can:

- onboard and manage restaurant partners and riders,
- monitor all orders and resolve disputes,
- view platform-wide analytics and revenue reports,
- and manage customer accounts.

All admin and partner pages are protected using role-based access control and session authentication.

The Customer Journey

The FoodFlow customer journey is structured as a five-stage funnel, designed to minimise drop-off at each step and guide the user from discovery to completed delivery:

Table 2 Customer Journey Stages

Stage	Phase	User Experience on FoodFlow
1	Discovery	User opens the app or website, views featured restaurants, promotional banners, and browses cuisine categories or uses the search bar to find specific dishes.
2	Consideration	User views a restaurant page, reads dish descriptions, checks prices, preparation time, and reviews left by other customers.
3	Intent	User adds selected items to the cart, chooses portion size or add-ons, and reviews the cart summary before proceeding.
4	Purchase	User selects a delivery address, chooses a payment method, confirms the order, and receives a real-time order confirmation with estimated delivery time.
5	Retention	User tracks order status live on the map, rates the delivery and meal after completion, and can reorder from order history with a single tap.

Technology and Platform Architecture

FoodFlow is built as a server-side rendered web application following a structured MVC-inspired layering approach within a PHP-native environment:

Presentation Layer: All HTML pages are rendered using semantic HTML5 markup and styled with custom CSS3, with no reliance on UI frameworks such as Bootstrap or Tailwind, in strict adherence to the assignment constraint.

Application Layer: PHP scripts handle all server-side logic, including form validation, session management, role enforcement, and database query execution. Common page components such as headers, footers, and navigation bars are included using PHP's require/include statements to promote code reusability.

Data Layer: MySQL serves as the relational database management system. The database is structured with normalised tables covering users, restaurants, menus, orders, order items, riders, and delivery records. All database interactions use prepared statements to prevent SQL injection attacks.

Security Layer: Client-side validation using JavaScript provides immediate user feedback, while PHP-based server-side validation acts as the authoritative data integrity check. Output is sanitised before rendering to prevent XSS vulnerabilities.

Business Differentiation

FoodFlow differentiates itself from existing food delivery and ordering platforms in three key ways:

Unified Experience: Unlike ordering through phone calls or social media pages, every part of FoodFlow — from browsing to payment to delivery tracking — is handled within a single platform, providing a complete and consistent experience.

Partner Empowerment: Restaurant partners are given meaningful tools to manage their own menus, availability, and order flow without needing third-party POS systems or separate management software.

Data Ownership: All customer, order, restaurant, and delivery data is stored within FoodFlow's own MySQL database, giving the business full visibility and control over its operational data without sharing insights with external platform operators.

Identify Target Users and Analyse Their Needs

FoodFlow serves four primary user groups. Their needs directly shaped the system's functional and non-functional requirements:

Table 3 Target User Groups and Their Needs

User Group	Profile	Key Needs and Expectations
Customers / Diners	Tech-savvy individuals aged 18–45. Students, working professionals, and families who prefer convenient online food ordering.	Fast restaurant search and filtering; clear menu with images and prices; transparent delivery tracking; secure payment options; order history and reorder features.
Restaurant Partners	F&B business owners and managers responsible for managing their menus and incoming orders on the platform.	Simple menu upload and update tools; real-time order notification system; order acceptance and rejection controls; sales and revenue summary reports.
Delivery Riders	Freelance or contracted delivery personnel who fulfil customer orders.	Clear order pickup instructions; integrated navigation support; earnings tracking dashboard; simple status update controls (Picked Up, On the Way, Delivered).
Admin / Platform Manager	Internal staff responsible for managing platform content, accounts, and operational oversight.	Full CRUD control over restaurants and menus; customer and rider account management; dispute resolution tools; platform-wide sales and analytics reports.

User Pain Points Analysis

Many customers and restaurant owners face difficulties when ordering or managing food delivery through existing channels. These issues negatively affect order accuracy, customer trust, and operational efficiency.

Common Customer Problems:

- Lack of real-time order tracking after payment
- Inconsistent menu information and outdated pricing
- No centralised order history for repeat ordering
- Confusing or insecure payment processes
- Limited customer support channels for delivery issues

Restaurant Partner Operational Problems:

- Manual order taking via phone with high error rates
- No structured digital menu management system
- Difficulty tracking daily sales and revenue
- Lack of rider coordination tools

FoodFlow aims to reduce these problems through an integrated and centralised food delivery management system.

Competitor Comparison

Instead of replicating the complexity of large-scale delivery ecosystems, FoodFlow focuses on a simpler, more transparent, and locally-focused food delivery experience.

Table 4 Competitor Comparison

Feature	Phone-based Orders	Social Media Ordering	Generic Delivery Apps	FoodFlow
Structured Restaurant Listing	No	No	Partial	Yes
Real-Time Delivery Tracking	No	No	Yes	Yes
Secure Login System	No	No	Yes	Yes
In-App Payment Processing	No	No	Yes	Yes
Order History & Reorder	No	No	Partial	Yes
Restaurant Dashboard	No	No	Limited	Yes
Rider Earnings Management	No	No	Limited	Yes
Brand Consistency	Weak	Weak	Moderate	Strong

Discuss Existing Challenges or Gaps the Proposed System Aims to Address

Based on the analysis of the Malaysian food delivery landscape and the broader online ordering ecosystem, the following key challenges have been identified — all of which FoodFlow is designed to resolve:

Table 5 Existing Challenges and How FoodFlow Addresses Them

Challenge / Gap	How FoodFlow Addresses It
Fragmented multi-platform ordering experience	FoodFlow consolidates all restaurants, menus, ordering, payment, and tracking into a single unified platform, removing the need for customers to use multiple apps or websites.

Lack of real-time delivery tracking for customers	An integrated live order tracking system displays order status from preparation through pickup and delivery, giving customers full visibility at every stage.
Poor restaurant and menu management tools	Restaurant partners access a dedicated dashboard for menu uploads, real-time order notifications, and sales reports, reducing reliance on manual processes.
Security vulnerabilities in small food ordering systems	FoodFlow implements prepared statements to prevent SQL Injection, output sanitisation to prevent XSS, and role-based session management to prevent unauthorised access.
Inefficient rider coordination and dispatch	Riders receive structured job assignments with integrated navigation support and a simple status update system, reducing miscommunication and late deliveries.
Absence of consolidated order history and analytics	Customers view complete order histories with reorder functionality; admins access platform-wide sales analytics and reports for data-driven decision making.

Business Value of the Proposed System

FoodFlow improves both customer experience and business operations by providing a centralised and organised food delivery platform. Customers can easily discover restaurants, browse full menus with images, manage their cart, pay securely, and track deliveries in real time. This creates a smoother, more reliable, and more satisfying food ordering experience.

For restaurant partners, the system simplifies menu management, order reception, and daily sales monitoring through a dedicated partner dashboard. This reduces dependency on phone-based orders and manual record keeping.

In addition, FoodFlow helps both the platform operator and its restaurant partners build stronger digital identities rather than relying entirely on social media or third-party aggregators. The system is designed to support future enhancements such as in-app payment gateway integration, customer loyalty programmes, advanced analytics, and AI-driven dish recommendations.

Proposed System Solution

Define System Objectives and Key Functionalities

The FoodFlow system is developed to provide a secure, organised, and user-friendly food delivery platform. The proposed system focuses on improving the customer ordering experience, enabling restaurant partner self-management, and streamlining delivery coordination through a centralised web application.

The main objectives of the system are:

- To provide customers with a fast and intuitive restaurant discovery and food ordering experience.
- To implement a secure authentication system with separate access levels for customers, restaurant partners, delivery riders, and administrators.
- To allow restaurant partners to manage menus, pricing, and incoming orders efficiently through a dedicated partner dashboard.
- To enable real-time order tracking for customers from order confirmation to doorstep delivery.
- To ensure data accuracy using both client-side and server-side validation across all forms.
- To improve system security using prepared statements, session management, and output sanitisation.
- To deliver a responsive and visually consistent interface using custom CSS without relying on external UI frameworks.

The system functionalities are divided into customer-facing, partner-facing, and admin-facing modules as described below.

Table 6 Order & Menu Management Functional Requirements

ID	Requirement	Version
F1.1	The system shall display a list of available restaurants with images, names, cuisine types, ratings, and estimated delivery time.	1.0
F1.2	The system shall allow restaurants to be filtered by cuisine category, price range, and rating.	1.0
F1.3	The system shall allow restaurants and dishes to be sorted by popularity, rating, or price.	1.0
F1.4	The user shall be able to search for restaurants or dishes using a keyword search bar.	1.0
F1.5	The system shall display detailed restaurant pages with full menus, dish images, descriptions, and prices.	1.0
F1.6	The system shall indicate when a menu item is unavailable or sold out.	1.0

Table 7 Account Management Functional Requirements

ID	Requirement	Version
F2.1	The user shall be able to register a new customer account with name, email, phone, and password.	1.0
F2.2	The user shall be able to log in using email and password.	1.0
F2.3	The user shall be able to log out securely, terminating the active session.	1.0
F2.4	The user shall be able to view and update profile information and saved delivery addresses.	1.0
F2.5	The system shall maintain user login sessions securely with expiration controls.	1.0

F2.6	Guest users shall be able to browse restaurants and menus without logging in.	1.0
F2.7	Guest users shall be required to log in before placing an order or proceeding to checkout.	1.0
F2.8	The system shall prevent duplicate account registration using the same email address.	1.0
F2.9	The system shall validate registration data on both client-side and server-side.	1.0

Table 8 Shopping Cart Functional Requirements

ID	Requirement	Version
F3.1	The user shall be able to add menu items to the shopping cart from the restaurant page.	1.0
F3.2	The user shall be able to view all cart items including dish name, quantity, add-ons, and subtotal.	1.0
F3.3	The user shall be able to update item quantities or selected add-ons within the cart.	1.0
F3.4	The user shall be able to remove individual items from the cart.	1.0
F3.5	The cart total shall update dynamically using JavaScript without requiring a page reload.	1.0

Table 9 Order Placement and Tracking Functional Requirements

ID	Requirement	Version
F4.1	The user shall be able to proceed to checkout from the shopping cart page.	1.0
F4.2	The user shall be able to enter or select a saved delivery address during checkout.	1.0
F4.3	The user shall be able to select a payment method during checkout.	1.0
F4.4	The system shall validate all checkout form data before order submission.	1.0
F4.5	The system shall generate an order confirmation page with order summary and estimated delivery time.	1.0
F4.6	The user shall be able to track order status in real time from the order tracking page.	1.0
F4.7	The user shall be able to view order history and reorder from previous orders.	1.0
F4.8	The system shall store all order details and delivery records in the database.	1.0

Table 10 Admin & Partner Management Functional Requirements

ID	Requirement	Version
F5.1	The admin and partner shall log in through a secure role-restricted login page.	1.0

F5.2	The restaurant partner shall be able to add, edit, and delete menu items.	1.0
F5.3	The restaurant partner shall be able to accept or reject incoming customer orders.	1.0
F5.4	The restaurant partner shall be able to view daily order summaries and revenue reports.	1.0
F5.5	The admin shall be able to onboard and manage restaurant partner accounts.	1.0
F5.6	The admin shall be able to assign and manage delivery rider accounts.	1.0
F5.7	The admin shall be able to view and manage all platform orders and resolve disputes.	1.0
F5.8	The system shall prevent non-admin and non-partner users from accessing restricted dashboards.	1.0
F5.9	The admin dashboard shall display platform-wide order, revenue, and user activity summaries.	1.0

Non-Functional Requirements

Besides functional features, the system also includes several non-functional requirements to ensure quality, usability, and security.

Table 11 Non-Functional Requirements

ID	Requirement
N1	The system shall provide a responsive interface compatible with desktop, tablet, and mobile devices.
N2	The system shall load restaurant listings and menus within an acceptable response time under normal network conditions.
N3	User passwords shall be hashed and all sessions securely managed with expiration controls.
N4	The system shall prevent unauthorised access to restaurant partner and admin pages using role-based access control.
N5	The system shall maintain data consistency and integrity for orders, payments, and delivery records within the database.
N6	The system codebase shall follow a modular and reusable structure to support maintainability and future enhancements.

Table 12 Customer & Partner Feature Summary

Customer-Facing Features	Admin & Partner-Facing Features
Restaurant and menu search with category filtering	Restaurant partner registration and menu management
Detailed restaurant and dish pages with images	Order management and real-time status updates

Shopping cart with quantity and add-on management	Rider assignment and delivery tracking oversight
User registration and secure login system	Admin dashboard with business summary reports
Checkout with delivery address and payment selection	Role-based access control for partner and admin users
Real-time order tracking map	Platform-wide sales analytics and revenue reporting
Order history and one-tap reorder	Customer account monitoring and dispute management

System Workflow

The FoodFlow system follows a structured workflow designed to guide customers from discovery through to successful order completion and delivery.

- The user browses or searches for restaurants through the homepage or restaurant listing page.
- The user selects a restaurant, views the menu, and adds desired items to the shopping cart.
- The user logs in or registers before proceeding to checkout.
- Delivery address and payment method are entered and validated during the checkout process.
- The restaurant partner receives and confirms the order through the partner dashboard.
- A rider is assigned and picks up the order; the customer tracks delivery status in real time.
- Upon delivery, the customer is prompted to rate the meal and the delivery experience.
- Administrators monitor platform operations, manage accounts, and review analytics through the admin dashboard.

How the Proposed System Improves Business Operations and User Experience

For customers, FoodFlow provides a smoother and more organised food ordering experience through clear restaurant categorisation, fast dish searching, real-time delivery tracking, and a simplified checkout process. Features such as dynamic cart updates, saved addresses, and one-tap reordering improve convenience and encourage repeat purchases.

For restaurant partners, the system centralises menu management, order reception, and revenue reporting within a single partner dashboard. This reduces dependency on phone-based ordering and manual record keeping, and helps partners make faster, more informed decisions.

From a business perspective, FoodFlow helps establish a stronger platform brand identity by providing a dedicated food delivery storefront. In addition, the modular system structure supports future enhancements such as in-app payment gateway integration, a loyalty and rewards programme, AI-driven dish recommendations, and advanced delivery optimisation.

Future Enhancements

The system is designed to support future improvements and scalability. Possible future enhancements include:

- Online payment gateway integration (e.g. FPX, credit/debit card, e-wallet support)
- Customer loyalty and rewards programme
- Dish rating and review system
- AI-powered dish and restaurant recommendation engine
- Email and push notifications for order status updates
- Advanced sales analytics and delivery performance reporting

System Design and Modelling

Develop System Design Using Wireframes or User Interface Mockups

The FoodFlow system is designed using wireframes and user interface mockups to clearly show the structure, layout, and navigation flow of the web application. The design focuses on simplicity, consistency, and ease of use for customers, restaurant partners, and administrators.

The system is divided into four access levels: public users, registered customers, restaurant partners, and administrators. Each access level has different pages and permissions based on the user role.

Table 13 System Site Map by Access Level

Access Level	Page / Screen	PHP File
Public	Home / Landing Page	index.php
Public	Restaurant Listing	restaurants.php
Public	Restaurant & Menu Detail	restaurant_detail.php
Public	User Registration	register.php
Public	User Login	login.php
Public	About / Contact	about.php
Customer	Shopping Cart	cart.php
Customer	Checkout	checkout.php
Customer	Order Confirmation	order_confirm.php
Customer	Order History & Tracking	order_history.php
Customer	My Profile	profile.php
Restaurant Partner	Partner Dashboard	partner/dashboard.php
Restaurant Partner	Manage Menu	partner/menu.php

Restaurant Partner	Incoming Orders	partner/orders.php
Admin	Admin Dashboard	admin/dashboard.php
Admin	Manage Restaurants	admin/restaurants.php
Admin	Manage Users & Riders	admin/users.php
Admin	Manage Orders	admin/orders.php
Admin	Analytics & Reports	admin/reports.php

UI/UX Design Principles

The interface design of FoodFlow follows several usability principles to ensure that users can interact with the system easily and efficiently.

First, the system uses a consistent layout across all customer-facing pages. The header, navigation bar, search bar, cart icon, and login button remain in the same position to make navigation familiar for users. This reduces confusion and improves the overall browsing experience.

Second, the design uses clear visual hierarchy. Important actions such as 'Order Now', 'Add to Cart', and 'Proceed to Checkout' are displayed as prominent call-to-action buttons so users can quickly understand the next step.

Third, restaurant and dish cards are designed to show the most important information clearly — restaurant name, cuisine type, rating, estimated delivery time, and minimum order — to help customers make faster decisions.

Fourth, the system applies a responsive layout so that all pages can adjust to different screen sizes, supporting customers who access the platform through laptops, tablets, or mobile devices.

Finally, the partner and admin interfaces are designed to be practical and task-focused. Dashboards provide quick access to key operational data such as incoming orders, menu items, revenue, and delivery status.

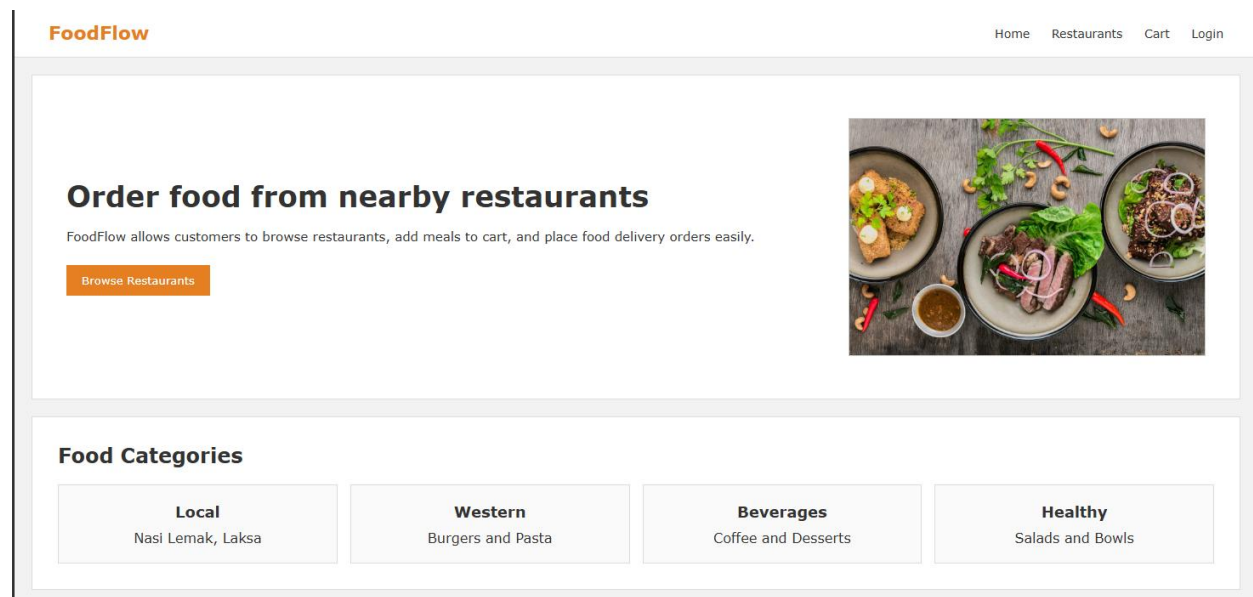
Home Page Design

The Home Page acts as the main entry point of the FoodFlow platform. It introduces the brand and helps users quickly start browsing restaurants and dishes.

The page includes a full-width header with the FoodFlow logo, location selector, search bar, cart icon, and login button. Below the header, the navigation bar provides links to major food categories such as All Restaurants, Local, Western, Beverages, Healthy, and Groceries.

The hero section contains the main tagline, 'Craving Something? We Deliver.', with call-to-action buttons such as 'Order Now' and 'Browse Restaurants'. This section is designed to attract users and guide them toward food discovery.

Below the hero section, the page includes a 'Browse by Category' area with cuisine cards and a 'Featured Restaurants' section displaying restaurant cards with images, ratings, estimated delivery times, and minimum order amounts.



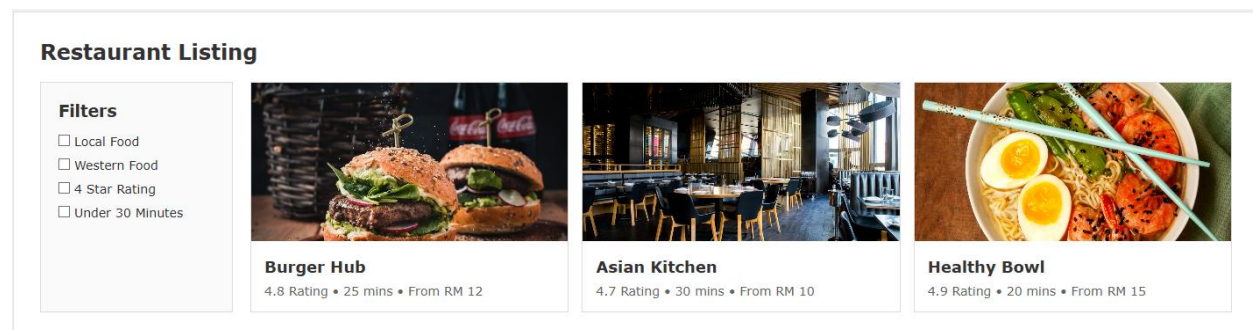
[Figure 1 - Home Page Wireframe]

Restaurant & Menu Page Design

The Restaurant Listing Page allows users to browse and filter all available restaurants. It uses a two-column layout.

The left sidebar contains filter options such as cuisine type, price range, rating, and delivery time. This helps users narrow down restaurants based on their preferences and location.

The main content area displays restaurants in a responsive grid layout. Each restaurant card includes the restaurant image, name, cuisine type, rating, estimated delivery time, and a minimum order indicator. The page also includes sorting options and pagination to improve browsing usability.



[Figure 2 - Restaurant Listing Page Wireframe]

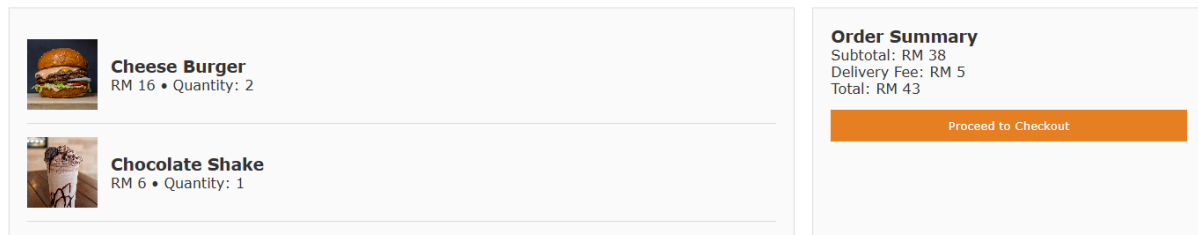
Shopping Cart Page Design

The Shopping Cart Page allows customers to review selected items before checkout.

Each cart item displays the dish image, dish name, selected add-ons, unit price, quantity controls, and total item price. Users can increase or decrease item quantities using JavaScript-controlled buttons and can remove items from the cart entirely.

The right side of the page contains an Order Summary section showing subtotal, delivery fee, applicable promotions, and the final total. It also includes a promo code field and a clear 'Proceed to Checkout' button.

Shopping Cart



The wireframe shows a shopping cart layout. On the left, there is a list of items. The first item is a 'Cheese Burger' with a price of RM 16 and a quantity of 2. The second item is a 'Chocolate Shake' with a price of RM 6 and a quantity of 1. On the right, there is an 'Order Summary' section. It displays the subtotal as RM 38, the delivery fee as RM 5, and the total as RM 43. Below the summary is an orange button labeled 'Proceed to Checkout'.

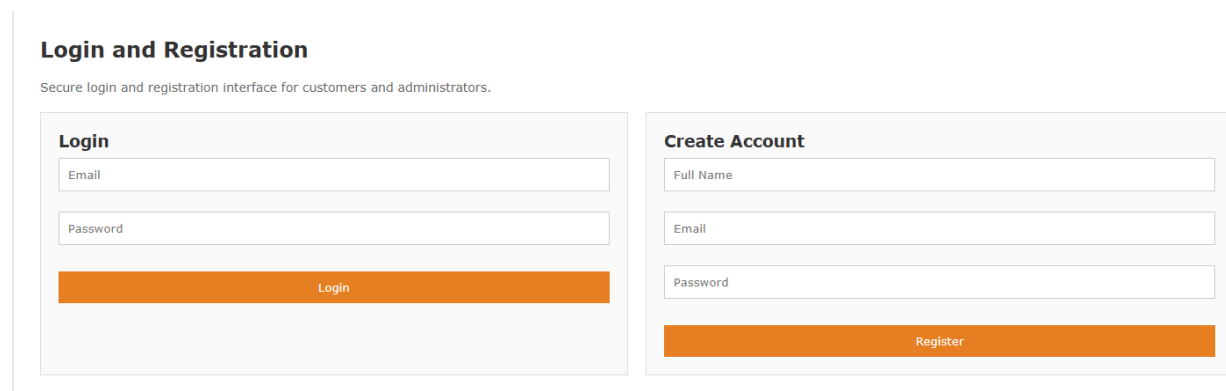
[Figure 3 - Shopping Cart Wireframe]

Login and Registration Page Design

The Login and Registration Page combines both sign-in and account creation forms in a single screen, making it easier for new and returning users to access the system.

The sign-in form allows existing users to log in using their email address and password, with 'Remember Me' and 'Forgot Password' options included.

The registration form allows new users to create an account by entering their full name, email address, phone number, password, and password confirmation, along with acceptance of the terms and conditions.



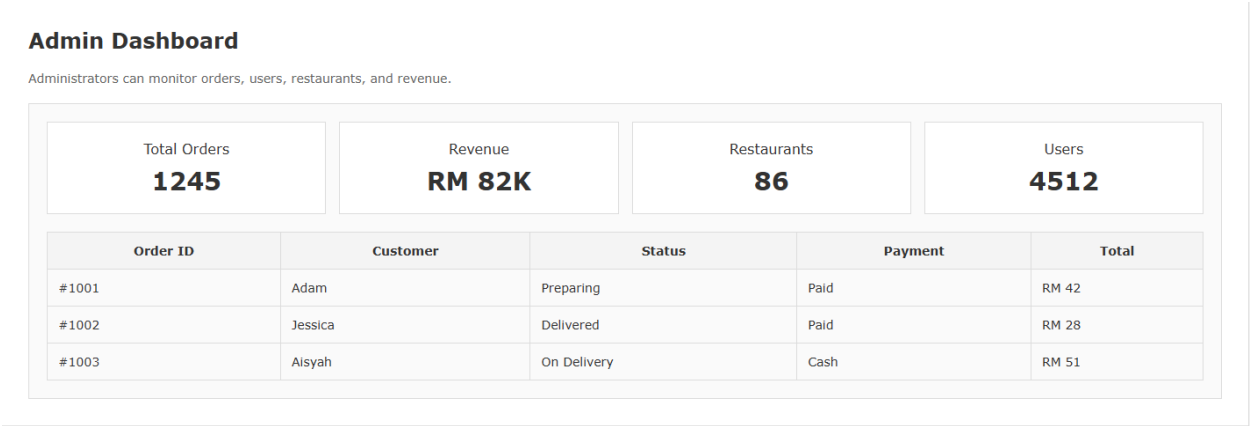
The wireframe shows a login and registration page. It has a header 'Login and Registration' with a subtitle 'Secure login and registration interface for customers and administrators.' Below the header, there are two main sections. The 'Login' section on the left has input fields for 'Email' and 'Password', and an orange 'Login' button. The 'Create Account' section on the right has input fields for 'Full Name', 'Email', and 'Password', and an orange 'Register' button.

[Figure 4 - Login and Registration Page Wireframe]

Admin Dashboard Design

The Admin Dashboard is designed for platform administrators to manage the full operations of FoodFlow.

The page uses a sidebar layout with navigation links including Dashboard, Manage Restaurants, Manage Riders, Manage Customers, Manage Orders, Analytics, and Settings. The main dashboard area displays summary cards for total orders, total revenue, active restaurants, and registered customers. A Recent Orders table shows order ID, customer name, restaurant, total amount, delivery status, and action controls. The dashboard also includes a Low Stock or Unavailable Menu Alert section to help administrators flag restaurants with issues requiring attention.



[Figure 5 - Admin Dashboard Wireframe]

Summary

The system design of FoodFlow focuses on a clean, modern, and user-friendly interface that serves all four user roles effectively. Customer-facing pages support intuitive restaurant discovery, cart management, and order tracking. Partner-facing pages support efficient menu and order management. Admin pages provide full platform oversight and operational analytics. Overall, the wireframes and mockups demonstrate that the proposed system follows consistent layout design, clear navigation, responsive structure, and practical usability principles that prioritise both the customer experience and business operations.

Project Planning and Management

Define Roles and Responsibilities of Each Group Member

The development of FoodFlow is divided among the group members based on system modules and responsibilities. Each member is assigned a minimum of three pages and three major features for individual assessment purposes.

Table 14 Roles and Responsibilities of Group Members

Member	Role	Pages Developed	Individual Features
Member Name	Role	Pages Developed	Individual Features

Al-Maweri Yousef Mohammed Abdullah	Lead Full-Stack Developer (Customer Experience)	Home Page (index.php), Restaurant Listing (restaurants.php), Restaurant Detail (restaurant_detail.php), Shopping Cart (cart.php), and Checkout Page (checkout.php).	• Hero section, category navigation, and responsive restaurant search/filtering . • Dynamic Menu display with dish images and pricing . • Shopping Cart CRUD with add-on/quantity controls and dynamic JS total updates . • Checkout processing and order confirmation logic.
Bara Samih Jamal Yousef	Lead Systems & Operations Developer (Admin/Security)	Register Page (register.php), Login Page (login.php), User Profile (profile.php), Admin Dashboard (admin/dashboard.php), and Partner Dashboard (partner/dashboard.php).	• Secure session-based authentication (Login/Logout) and multi-step registration with PHP validation . • Admin CRUD for restaurants, users, and platform-wide order oversight . • Partner dashboard for menu management and real-time order acceptance . • Lead Architect: Database schema design (ERD), SQL prepared statements, and security implementation (XSS/SQLi protection).

Development Methodology

The FoodFlow project follows an Agile-inspired development approach where the system is developed incrementally through multiple phases. Tasks are distributed among team members based on modules and responsibilities, while GitHub is used for collaboration and version control.

Weekly progress discussions are conducted to monitor development progress, resolve integration issues, and ensure consistency between frontend and backend modules.

Project Risk Management

Several potential project risks have been identified during the planning phase. Appropriate mitigation strategies are prepared to reduce development issues and delays.

Table 15 Project Risks and Mitigation Strategies

Risk	Mitigation Strategy
Merge conflicts during team collaboration	Use GitHub branching and conduct regular code synchronisation sessions
Delays in module completion	Conduct weekly progress reviews and task monitoring with assigned deadlines
Database connection or integration issues	Perform continuous backend testing throughout each development phase
UI inconsistency across modules	Follow shared design guidelines, colour tokens, and agreed wireframe mockups
Security vulnerabilities in authentication and payment flows	Apply validation, prepared statements, session protection, and output sanitisation
Feature integration conflicts between modules	Test modules incrementally and perform full integration testing before final submission

Identify Tools, Frameworks, and Technologies to Be Used

The development of FoodFlow uses several tools and technologies to support frontend design, backend development, database management, collaboration, and testing.

Table 16 Tools, Frameworks, and Technologies

Category	Technology / Tool
Frontend	HTML5, CSS3 (custom styling only), Vanilla JavaScript
Backend	PHP for server-side logic, session management, and validation
Database	MySQL with PDO prepared statements
Server Environment	XAMPP (Apache, PHP, MySQL)
Version Control	Git and GitHub
UI Design & Mockups	Figma

Code Editor	Visual Studio Code
Security Implementation	PHP Validation, Session Authentication, Prepared Statements, Output Sanitisation

Project Timeline

The project timeline outlines the planned development phases and major deliverables for the FoodFlow system.

Table 17 Project Timeline

Week	Date	Phase	Milestone / Deliverable
Week 7	11 May 2026 – 17 May 2026	Project Initiation	Finalise project scope, assign modules, and begin database and UI planning
Week 8	18 May 2026 – 24 May 2026	System Design & Planning	Complete UI mockups, ERD design, and database schema planning
Week 9	25 May 2026 – 31 May 2026	Frontend Development	Develop customer-facing pages using HTML5, CSS3, and JavaScript
Week 10	1 June 2026 – 7 June 2026	Backend Development	Implement PHP backend modules, authentication, and database connectivity
Week 11	8 June 2026 – 14 June 2026	System Integration & Testing	Integrate all modules, perform validation, security, and usability testing
Week 12	15 June 2026 – 21 June 2026	Proposal & Part 1 Submission	Finalise documentation, fix remaining issues, and submit Part 1 before 21 June 2026
Week 13	22 June 2026 – 28 June 2026	Improvement & Refinement	Improve system functionality and resolve feedback from supervisor
Week 14	29 June 2026 – 5 July 2026	Individual Interview	Prepare for individual evaluation and project demonstration sessions